

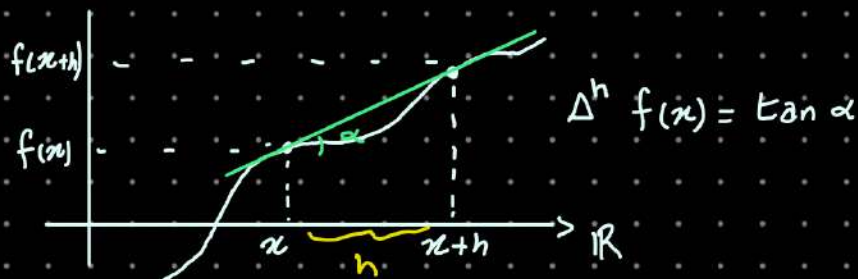
Derivale e calc. diff

Sia $f: (a,b) \rightarrow \mathbb{R}$ e sia $x \in (a,b)$.

Definiamo il **rapporto incrementale** (di h , rispetto a x):

$$\frac{f(x+h) - f(x)}{x+h-x}, \quad h \neq 0$$

e anche $|h|$ piccolo in modo che $x+h \in (a,b)$.



Def Sia $f: (a,b) \rightarrow \mathbb{R}$

Si defin **DERIVATA** di f **NEL P.TO** x

il numero:

$$f'(x) := \lim_{\substack{h \rightarrow 0 \\ (h \neq 0)}} \frac{f(x+h) - f(x)}{h}$$

Perché il limite **ESISTA** e sia **FINITO**

- Se $f'(x)$ è defn. diremo che f è **DERIVABILE** nel p.to x
- Se f è derivabile $\forall x \in (a,b)$, diremo che f è deriv. in (a,b)

In questo caso, la funz.

f'

Altre notaz.

- Df
- $\frac{df}{dx}$
- $\dot{f}$

il signif. di f' è quello di "tasso di variazione istantaneo" di f rispetto a x

Esempio

- Se " $t \in \mathbb{R}$ " è il tempo e
 $S: \mathbb{R} \rightarrow \mathbb{R}$ è una posizione spaziale
 - $t \in \mathbb{R}$ è il tempo
 $S: \mathbb{R} \rightarrow \mathbb{R}$ è una velocità
 $\frac{ds}{dt}$ è l'accelerazione.
-

Se f è derivabile in x

$$\Delta^h f(x) = m_{\alpha_h} = \tan \alpha_h$$

$$\downarrow$$
$$f'(x) = \tan(\alpha)$$

per defin.

$$f'(x) = \lim_{h \rightarrow 0} \underbrace{\Delta^h f(x)}_{\substack{\text{incremento} \\ \text{nella} \\ \text{dim di } h}} = \lim_{h \rightarrow 0} \underbrace{m_{\alpha_h}}_{\text{cost.}} = \lim_{h \rightarrow 0} \tan(\alpha_h) = \tan(\lim_{h \rightarrow 0} \alpha_h)$$

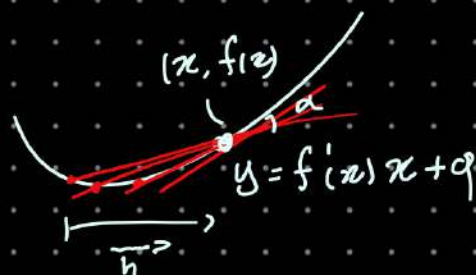
$$\begin{aligned} \lim_{h \rightarrow 0} \alpha_h &= \lim_{h \rightarrow 0} \tan^{-1}(m_h) \\ &= \tan^{-1}(\lim_{h \rightarrow 0} m_h) \\ &= \tan^{-1}(f'(x)) \end{aligned}$$

$$f(x) = f'(x) + q$$

Vogliamo che $(x_0, f(x_0))$ sia un p.to

$$\text{in } r = " \lim r_h "$$

$$\begin{aligned} y &= f'(x) (x - x_0) + y_0 \\ &= f'(x) (x - x_0) + f(x_0) \end{aligned}$$



Se x_0 è dato e f è derivabile in x_0 , allora esiste una retta " $y = mx + q$ " tg al grafico di f nel pto $(x_0, f(x_0))$

Inoltre, questa è

$$y = f(x_0) + f'(x_0)(x - x_0)$$

Ricordiamo che

$$\lim_{x \rightarrow x_0} g(x) \text{ esiste finito}$$

Se e solo se

$$\begin{cases} \lim_{x \rightarrow x_0^+} g(x) \\ \lim_{x \rightarrow x_0^-} g(x) \end{cases}$$

• Allora $f'(x_0)$ esiste

$$e \quad f'(x_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

esistono, sono finiti e uguali

- f'_+ è la der. destra
- f'_- è la der. sinistra

ES

$$f(x) = |x|$$



$$\begin{cases} f' \text{ esiste in } (0, +\infty) = 1 \\ f'_+ // \text{ in } (-\infty, 0) = -1 \\ f'_+ // \text{ in } x=0, = 1 \\ f'_- // \text{ in } x=0, = -1 \end{cases}$$

però $f'(0)$ non esiste ($1 \neq -1$)

ES $f: \mathbb{R} \rightarrow \mathbb{R}$

Sia $f(x) = x^2$

Calcoliamo

$$\begin{aligned} f(x+h) - f(x) &= (x+h)^2 - x^2 \\ &= x^2 + 2xh + h^2 - x^2 \\ &= 2xh + h^2 \end{aligned}$$

$$\begin{aligned} \Delta^h f(x) &= \frac{f(x+h) - f(x)}{h} \\ &= \frac{2xh + h^2}{h} \\ &= 2x + h \end{aligned}$$

Dunque

$$\begin{aligned} f'(x_0) &= (x^2)'|_{x_0} \\ &= \lim_{h \rightarrow 0} (2x_0 + h) \\ &= 2x_0 \quad \forall x_0 \in \mathbb{R} \end{aligned}$$

$$\Rightarrow (x^2)' = 2x$$

Der. di potenza n

Sia $n \in \mathbb{N}$, ($n \geq 1$) e $f(x) = x^n$

$$\underline{f(x+h) - f(x)}$$

$$f(x+h) = (x+h)^n$$

bin. di Newton $= x^n + \binom{n}{1} x^{n-1} h + \binom{n}{2} x^{n-2} h^2$

$$\boxed{\binom{n}{k} = \frac{n!}{k!(n-k)!}}$$

$$+ \dots + \binom{n}{k} x^{n-k} h^k + \dots$$

$$\binom{n}{n-1} x^1 h^{n-1} + \binom{n}{n} h^n$$

Allora

$\nearrow$ Per cancellare

$$f(x+h) - f(x) = \binom{n}{1} x^{n-1} h + \dots + \binom{n}{k} x^{n-k} h^k + \dots + \binom{n}{n} h^n$$

$$\lim_{h \rightarrow 0} \Delta^h f(x) = \lim_{h \rightarrow 0} \frac{\binom{n}{1} x^{n-1} h + \dots + \binom{n}{n} h^n}{h}$$

$$= \binom{n}{1} x^{n-1} \cdot \lim_{h \rightarrow 0} \frac{h}{h} \cdot 1$$

$$f'(x) = \binom{n}{1} x^{n-1}$$

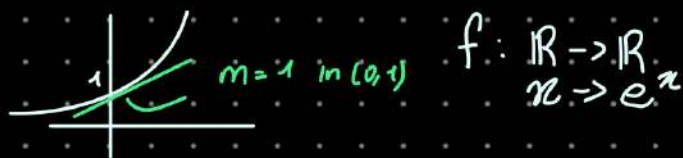
$$= \frac{n!}{1! (n-1)!} = \frac{1 \cdot 2 \cdot 3 \dots n}{1 \cdot \cancel{1} \cdot 2 \cdot 3 \dots n-1}$$

$$\Rightarrow f'(x) = n x^{n-1}$$

Es

$$x^3 = 3x^2$$

LA DERIV. delle esponenz.



abbiamo visto che

$$1 = \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= (e^x)'(0)$$

Sia $x \in \mathbb{R}$

$$\frac{f(x+h) - f(x)}{h} = \frac{e^{x+h} - e^x}{h}$$

$$= \frac{e^x \cdot e^h - e^x}{h}$$

$$= e^x \left(\frac{e^h - 1}{h} \right)$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = e^x \cdot \lim_{h \rightarrow 0} \frac{e^h - 1}{h} \cdot 1$$

$$= e^x \cdot 1$$

Dunque

$$(e^x)' = e^x$$

Continuità delle funz. derivabili

proposiz. Sia $f: (a, b) \rightarrow \mathbb{R}$ se f è derivab. in $x_0 \in (a, b)$ allora f è continua in x_0

Dim. per $h \neq 0$, $x_0 + h \in (a, b)$
si ha:

$$\begin{aligned} f(x_0 + h) - f(x_0) &= 1 \cdot [f(x_0 + h) - f(x_0)] \\ &= h \cdot \frac{f(x_0 + h) - f(x_0)}{h} \\ &\quad \downarrow \quad \quad \quad \searrow \text{ } f'(x_0) \in \mathbb{R} \\ &\quad \quad \quad \underbrace{\quad \quad \quad}_{0 \cdot f'(x_0) = 0} \end{aligned}$$

$$f(x_0 + h) - f(x_0) \rightarrow 0$$

$$\begin{cases} y = x_0 + h \\ h \rightarrow 0 \end{cases} \Rightarrow y \rightarrow x_0$$

$$f(y) \rightarrow f(x_0) \quad \text{se } y \rightarrow x_0$$

$$\begin{aligned} |f(x+h) - f(x)| &= \underline{h} \cdot \left| \frac{f(x+h) - f(x)}{h} \right| \\ &\quad \searrow \text{ } f'(x) \end{aligned}$$

$$[f(x+h) - f(x)] - f'(x)h = o(h)$$

Es $f(x)$



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$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$f: (a, b) \rightarrow \mathbb{R} \quad x_0 \in (a, b)$$

DEF f DERIVABILE IN $x_0 \stackrel{\text{def}}{\iff} f'(x_0)$ è finito

DEF Limite

$$\{x_n\} \subset (a, b) \quad x_n \neq x_0 \quad \lim_{n \rightarrow +\infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} = f'(x_0)$$

PER OGNI $\{x_n\}$ $x_n \neq x_0$

DEF $\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in (a, b) \text{ t.c. } |x - x_0| < \delta \quad x \neq x_0$
 $\left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| < \varepsilon$



$$\frac{f(x) - f(x_0)}{x - x_0} \text{ è "well defined"}$$

TEOR f è derivabile in $x_0 \Rightarrow f$ è continua in x_0

OSS

non vale l'opposto



$f(x) = |x|$ è continua ma non deriv.

$$f(x) = \begin{cases} 1 & x < 1 \\ 2 & x = 1 \\ 3 & x > 1 \end{cases}$$

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

$$x_0 = 1$$

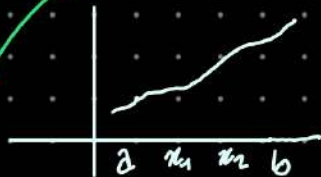
$$\frac{f(1+h) - f(1)}{h} = \frac{3-2}{h} = \frac{1}{h} \quad h > 0$$

Prop. incr. da dx

$$\lim_{h \rightarrow 0}$$

TEOR f **crescente**, f derivabile $\Rightarrow f'(x) \geq 0 \quad \forall x \in (a, b)$
 $f: (a, b) \rightarrow \mathbb{R}$

DIM f cresc $a < x_1 < x_2 < b \Rightarrow f(x_1) \leq f(x_2)$
 $\forall x_1, x_2 \in (a, b)$



$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} \geq 0 \quad x_1 < x_2$$

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} \geq 0 \quad x_1 > x_2$$

$$x_1 - x_2 > 0$$

$$f(x_1) \geq f(x_2)$$

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} \geq 0 \quad \forall x_1, x_2 \in (a, b)$$

$$\frac{f(x) - f(x_0)}{x - x_0} \geq 0 \quad \text{perché } f \text{ cresc.}$$

$$\Rightarrow \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \geq 0 \quad \text{per. teor. confronto.}$$

$$x_n \rightarrow x_0 \quad x_n \neq x_0 \rightarrow b_n = \frac{f(x_n) - f(x_0)}{x_n - x_0} \geq 0 \quad \lim_{n \rightarrow +\infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} \geq 0$$

TEOR f $\in$ **strett. crescente** + DERIVABILE $\Rightarrow f'(x_0) > 0$
 $f: (a, b) \rightarrow \mathbb{R}$

$$a < x_1 < x_2 < b \quad f(x_1) < f(x_2)$$

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} > 0$$

$$x_n \rightarrow x_0$$

$$x_n \neq x_0$$

$$\frac{f(x_n) - f(x_0)}{x_n - x_0} = b_n > 0$$

$$\lim b_n = L \Rightarrow L \geq 0$$

ES

$$f'(x_0) > 0 \quad x_0 \in (a, b)$$

$$0 < f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{n \rightarrow +\infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} > 0 \quad \begin{matrix} x_n \rightarrow x_0 \\ x_n \neq x_0 \end{matrix}$$

$$\text{TEO PERM SEGN} \quad b_n \rightarrow L > 0 \rightarrow \begin{matrix} b_n > 0 \\ \exists N: b_n > 0 \quad \forall n > N \end{matrix}$$

$$\frac{f(x_n) - f(x_0)}{x_n - x_0} \longrightarrow f'(x_0) > 0$$

$$\Rightarrow \frac{f(x_n) - f(x_0)}{x_n - x_0} > 0 \quad \text{DEFINITIVAM}$$

Regole di derivazione

$$\text{HP} \quad f, g \text{ derivabili in } x_0 \quad f, g: (a, b) \rightarrow \mathbb{R} \quad x_0 \in (a, b)$$

$$f + g \quad x \mapsto f(x) + g(x) = (f + g)(x)$$

$$f \cdot g \quad x \mapsto f(x) g(x) = (f \cdot g)(x)$$

$$\frac{f}{g} \quad x \mapsto \frac{f(x)}{g(x)} = \left(\frac{f}{g}\right)(x)$$

$$(f + g)'(x_0) = f'(x_0) + g'(x_0)$$

$$(f \cdot g)'(x_0) = f'(x_0) \cdot g(x_0) + f(x_0) \cdot g'(x_0)$$

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0) \cdot g(x_0) - f(x_0) \cdot g'(x_0)}{(g(x_0))^2} \quad \text{con } g(x_0) \neq 0$$

$$\text{DIM} \quad \frac{(f + g)(x) - (f + g)(x_0)}{x - x_0} = \frac{\underbrace{f(x) - f(x_0)}_{f'(x_0)} + \underbrace{g(x) - g(x_0)}_{g'(x_0)}}{x - x_0} =$$

$$(x + e^x)' = 1 + e^x$$

$$(f + g + h)' = [(f + g) + h]' = (f + g)' + h' = f' + g' + h'$$

$$c f(x) \quad c \in \mathbb{R}$$

$$c \frac{f(x) - f(x_0)}{x - x_0} = c \frac{f(x_0) - f(x_0)}{x - x_0} \xrightarrow{x \rightarrow x_0} c f'(x_0)$$

$$(c f(x))' = c f'(x)$$

$$(\alpha_1 f(x) + \alpha_2 g(x))' = (\alpha_1 f(x))' + (\alpha_2 g(x))' = \alpha_1 f'(x) + \alpha_2 g'(x) \quad \alpha_1, \alpha_2 \in \mathbb{R}$$

$$\frac{d}{dx} f = f'$$

PRODOTTO

$$\frac{(f \cdot g)(x) - (f \cdot g)(x_0)}{x - x_0} = \frac{f(x) - f(x_0)}{x - x_0} g(x) + f(x_0) \frac{g(x) - g(x_0)}{x - x_0} \quad x \rightarrow x_0$$

$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$ $g(x_0) + f(x_0) g'(x_0)$
 $f, g, \text{ derivab} \Rightarrow g \text{ cont.}$ $g \text{ e deriv}$

$$\frac{\frac{f}{g}(x) - \frac{f}{g}(x_0)}{x - x_0} \quad g(x_0) \neq 0$$

TEO PERM SEGUO per
funz continue $\Rightarrow g(x) \neq 0$
Almeno vicino a x_0

$$(x^n)' = n x^{n-1} \quad \text{dim per induzione}$$

Supponi che per $N \in \mathbb{N}$

$$\frac{d}{dx} x^N = N x^{N-1}$$

$$\begin{aligned} \frac{d}{dx} (x^{N+1}) &= \frac{d}{dx} (x^N \cdot x) = N x^{N-1} \cdot x + x^N \cdot 1 \\ &= (N+1) x^N \end{aligned}$$

$$P(x) = \sum_{k=0}^m a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m$$

$$\begin{aligned} \frac{d}{dx} P(x) &= P'(x) = \frac{d}{dx} (a_0 x^0 + \dots + a_m x^m) \\ &= a_1 + 2a_2 x + \dots + m a_m x^{m-1} \end{aligned}$$

$$P'(x) = \sum_{k=1}^m k a_k x^{k-1}$$

ES 2) $\frac{d}{dx} \sin(x) = \lim_{h \rightarrow 0} \frac{\sin(x_0+h) - \sin(x_0)}{h} \stackrel{\text{S10+cosi}}{=} \lim_{h \rightarrow 0} \sin(x_0) \frac{\cos h - 1}{h} + \cos x_0 \frac{\sin h}{h}$

$\Rightarrow (\sin x)' = \cos x$

$$(\cos x)' = -\sin x$$

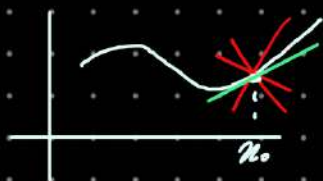
$$\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = \lim_{n \rightarrow \infty} \frac{\sin(h_n)}{h_n} = 1$$

$$h_n \rightarrow 0 \quad h_n \neq 0 \quad \forall \{h_n\}_{h_n \neq 0, h_n \rightarrow 0}$$

Es $\frac{d}{dx} \tan x = \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{\left(\frac{d}{dx} \sin x \right) \cos x - \sin x \left(\frac{d}{dx} \cos x \right)}{\cos^2 x}$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2 x} \begin{cases} \frac{1}{\cos^2 x} \\ 1 + \tan^2(x) \end{cases} \quad x \neq \frac{\pi}{2} + k\pi$$

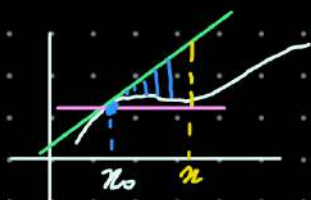
RETTA TANGENTE AL GRAFICO IN $(x_0, f(x_0))$



$$\psi(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$y - f(x_0) = m(x - x_0) \quad m = f'(x_0)$$

la retta tg in $(x_0, f(x_0))$ è la retta che "meglio approssima" $f(x)$ vicino a x_0



ψ è una retta passante per $(x_0, f(x_0))$
 f è continua

$$\psi(x) - f(x) \rightarrow 0 \quad x \rightarrow x_0$$

$$\psi(x) = m(x - x_0) + f(x_0)$$

$$\psi(x) - f(x) = m(x - x_0) + f(x_0) - f(x) \xrightarrow{x \rightarrow x_0} 0$$

$$\lim_{x \rightarrow x_0} \psi(x) - f(x) = 0 \quad \text{e} \quad \lim_{x \rightarrow x_0} \frac{\psi(x) - f(x)}{x - x_0} = 0$$

$\psi(x) = f(x_0) + m(x - x_0)$ Fascio di rette passante per $(x_0, f(x_0))$

$$\textcircled{1} \quad f(x_0) + m(x - x_0) - f(x) \rightarrow 0 \quad x \rightarrow x_0$$

$$\textcircled{2} \quad \frac{f(x_0) + m(x - x_0) - f(x)}{x - x_0} \xrightarrow{x \rightarrow x_0} 0$$

$$\frac{f(x_0) - f(x) + m(x - x_0)}{x - x_0} = \frac{f(x_0) - f(x)}{x - x_0} + \frac{m(x - x_0)}{x - x_0} = - \frac{f(x) - f(x_0)}{x - x_0} + m$$

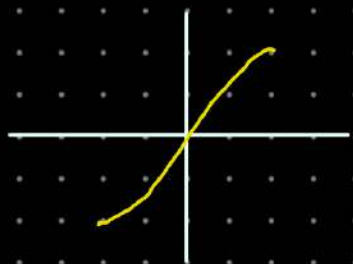
$$\downarrow \quad x \rightarrow x_0 \quad 0 \quad -f'(x_0) + m = 0$$

$$m = f'(x_0)$$

DEF RETTA TG

ψ retta TG È L'UNICA retta, passante per $(x_0, f(x_0))$ t.c.

$$\lim_{x \rightarrow x_0} \frac{\psi(x) - f(x)}{x - x_0} = 0 \quad \psi = f(x_0) + f'(x_0)(x - x_0)$$



$$f(x_0) \quad f(x) = \sin x \quad x=0$$

$$f(0) = \sin 0 = 0$$

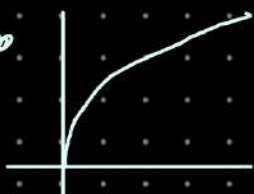
$$y = 1(x-0) + 0$$

$$f'(0) = \cos 0 = 1$$

$$f'(x_0)(x-x_0) + f(x_0)$$

$$\sin 1^\circ = ?$$

$$1^\circ = \frac{1}{180}$$



$$f(x) = \sqrt{x}$$

$$f(81) = 9$$

$$80 \sim 81$$

$$\varphi(x) = f(81) + f'(81)(x-81) = 9$$

$$f(x_0) + f'(x_0)(x-x_0)$$

$$\varphi(80) \approx f(80)$$

$$\lim_{x \rightarrow x_0} \frac{\sqrt{x} - \sqrt{x_0}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\sqrt{x} - \sqrt{x_0}}{(\sqrt{x} - \sqrt{x_0})(\sqrt{x} + \sqrt{x_0})} = \frac{1}{\sqrt{x_0} + \sqrt{x_0}} = \frac{1}{2\sqrt{x_0}}$$

$$f(x) = \sqrt{x} \quad f'(x) = \frac{1}{2\sqrt{x}}$$

$$D_f = \{x: x \geq 0\} \quad D_{f'} = \{x: x > 0\}$$

$$\varphi(80) = 9 + \frac{1}{18}(80-81) = 9 - \frac{1}{18}$$

- ① Rendere quantitativo
- ② Altre inform. geometriche (appross per eccesso o dif.?)

COMPOSIZ. DI FUNZIONI

$$f(x) = \sin(x^2)$$

$$x \mapsto x^2 \mapsto \sin x^2$$

$$f'(x) = ?$$

$$\lim_{x \rightarrow x_0} \frac{f(g(x)) - f(g(x_0))}{x - x_0} = \frac{d}{dx} f(g(x)) \big|_{x=x_0}$$

$$\frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} \cdot \frac{g(x) - g(x_0)}{x - x_0}$$

$$y = g(x)$$

$$y_0 = g(x_0)$$

$$\frac{f(y) - f(y_0)}{y - y_0} \cdot \frac{g(x) - g(x_0)}{x - x_0}$$

$$f'(y_0) \cdot g'(x_0) = f'(g(x_0)) \cdot g'(x_0)$$

Regola deriv f composta

$$(f(g(x)))' = f'(g(x)) g'(x)$$

$$\frac{d}{dx} \sin(x^2) = 2x \cos(x^2) \quad \hat{g(x)}$$

$$\frac{d}{dx} \sin\left(\frac{1}{x}\right) = \cos\left(\frac{1}{x}\right) \cdot \frac{d}{dx}\left(\frac{1}{x}\right) = \cos\left(\frac{1}{x}\right) \frac{d}{dx}(x^{-1}) = \cos\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right)$$

$$\lim_{x \rightarrow x_0} \frac{x - x_0}{x - x_0}$$

Eserc.

IMPORTANTE

$$f(x) = \begin{cases} x \cdot \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

I) f è continua?

① Se $x \neq 0$ $f(x) = x \cdot \sin\left(\frac{1}{x}\right)$

→ continua per comp. di f. cont

f è cont. per $x \neq 0$

$$\forall \varepsilon > 0 \exists \delta > 0: |x - 0| < \delta \quad |f(x) - f(0)| < \varepsilon \quad \stackrel{\text{def}}{=} f \text{ cont. in } x_0 = 0$$

$$f(x) = x \sin\left(\frac{1}{x}\right) \neq 0 \xrightarrow{\text{tende}} f(0) = 0$$

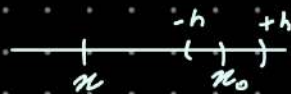
$$0 \leq |f(x) - f(0)| = \left| x \sin\left(\frac{1}{x}\right) - 0 \right| = \left| x \sin\left(\frac{1}{x}\right) \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| \xrightarrow{x \rightarrow 0} 0$$

f è cont. anche in $x = 0$

II) f è derivabile?

$$x_0 \neq 0$$

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$



$$f'(x) = 1 \cdot \sin\left(\frac{1}{x}\right) + x \cos\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) \quad x \neq 0$$

$$= \sin\left(\frac{1}{x}\right) - \frac{1}{x} \cos\left(\frac{1}{x}\right)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h \sin\left(\frac{1}{h}\right) - 0}{h} = \lim_{h \rightarrow 0} \sin\left(\frac{1}{h}\right) \quad \text{NE!}$$

ES PER CASA

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Svolgim

I) f cont.?

II) f deriv?
quanto vale $f'(x)$

28/10

$$\lim_{x \rightarrow x_0} f(x) = L$$

$$\textcircled{1} \quad \forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D(f) \quad |x - x_0| < \delta \quad x \neq x_0 \quad |f(x) - L| < \varepsilon$$

$$\textcircled{2} \quad \forall \{x_n\} \quad \{x_n\} \subset D(f) \quad x_n \neq x_0$$

$$\lim_{n \rightarrow +\infty} f_n = \lim_{n \rightarrow +\infty} f(x_n) = L$$

CAMBIO DI VARIABILE NEI LIMITI

$$\lim_{x \rightarrow x_0} g(x) = L$$

$$\lim_{x \rightarrow x_0} f(g(x)) = M$$

$$y = g(x)$$

$$\lim_{x \rightarrow L} f(x) = M$$

$$\lim_{y \rightarrow L} f(y) = M$$

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2} = \lim_{x \rightarrow 0} \frac{\sin(y)}{y} = 1$$

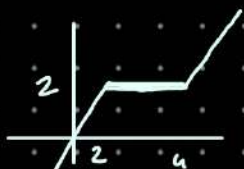
$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$y = \underbrace{e^x - 1}_0$$

$$e^x = 1 + y$$

$$x = \ln(1 + y)$$

$$\lim_{y \rightarrow 0} \frac{y}{\ln(1+y)} = 1$$



$$g(x) = \begin{cases} x+2 & x > 4 \\ 2 & x \in [2, 4] \\ x+2 & x < 2 \end{cases}$$

$$\lim_{x \rightarrow 3} g(x) = 2$$

$$f(x) = \begin{cases} 1 & x \neq 2 \\ 5 & x = 2 \end{cases}$$

$$\lim_{x \rightarrow 2} f(x) = 1$$



$$\lim_{x \rightarrow 3} f(g(x))$$

$$\lim_{y \rightarrow 2} f(y) = 1$$

$$y = g(x)$$

$$y \rightarrow 2$$

$$x \rightarrow 3$$

$\lim_{x \rightarrow 3} f(g(x)) = 5$
 $x \in (3 - \frac{1}{2}, 3 + \frac{1}{2})$
 $f(x) = 2$
 $f(g(x)) = 5$
 $x \in (3 - \frac{1}{2}, 3 + \frac{1}{2})$

$\lim_{x \rightarrow x_0} f(g(x))$
 $y = g(x)$
 $g(x) = y \rightarrow L$
 $= \lim_{y \rightarrow L} f(y)$

SE ① f è continua in $x = L$

SE ② $g(x) \neq L$

TEOREMA Fermat

$f: [a, b] \rightarrow \mathbb{R}$ $x_0 \in (a, b)$

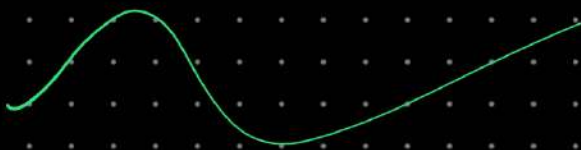
Supponiamo che

1) x_0 pto di min o max locale

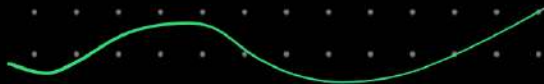
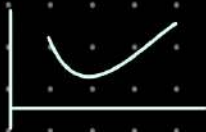
2) f derivab in x_0

$\Rightarrow f'(x_0) = 0$

non vale $\Leftarrow$



DIM



f dec $\rightarrow f' \leq 0$

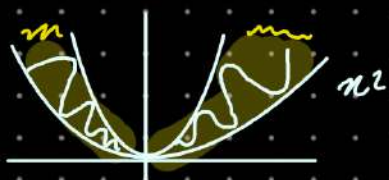
f cresc $\rightarrow f' \geq 0$

$\Rightarrow f'(x_0) = 0$

f esiste in

$$f(x) = \begin{cases} x^2 [\sin(\frac{1}{x}) + 2] & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$x^2 \leq x^2 (\sin(\frac{1}{x}) + 2) \leq 3x^2 \quad x \neq 0$$



$$f(0) = 0 < f(x) \quad \forall x \neq 0$$

$$f'(x) = \begin{cases} 2x (\sin(\frac{1}{x}) + 2) - \cos(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Su x "vicina" a zero $|2x (\sin(\frac{1}{x}) + 2)| < \frac{1}{2}$

Dim $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$

$$\frac{f(x) - f(x_0)}{x - x_0} \begin{matrix} \geq 0 \\ \leq 0 \end{matrix} \quad x < x_0 \quad f(x_0) \leq f(x) \quad (x_0 - \delta, x_0 + \delta) \\ f(x) - f(x_0) \geq 0$$

$$x_0 - \delta < x < x_0$$

$$\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \begin{matrix} \geq 0 \\ \leq 0 \end{matrix} \quad x > x_0$$

$$x_0 < x < x_0 + \delta$$

$$\lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \geq 0 \quad \text{Teo perim segno}$$

Quindi $f'(x_0) \geq 0$ e $f'(x_0) \leq 0 \Rightarrow f'(x_0) = 0$

$$f(x) = \begin{cases} x^2 & x \in \mathbb{R} \setminus \mathbb{Q} \\ 0 & x \in \mathbb{Q} \end{cases}$$



ES

ERRORE x_0 è pto min/max $\Rightarrow f'(x_0) = 0$

$$f(x) = \begin{cases} -1 & x=2 \\ 10 & x \neq 2 \end{cases}$$



$$f = |x|$$



$x_0 = 2$ pto min

$f'(x)$ N.E.

$$f(x) = x \quad x \in [1, 2] \quad f(1) = 1 \leq f(x) \quad \forall x \in (1, 2]$$



$$f'_+(x) = 1$$

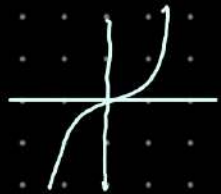
$$f'_-(1) \neq 0$$

"Ricetta" per cercare i possibili p.ti max/min

x_0 è p.to max/min relativo e f DERIVABILE $\Rightarrow f'(x_0)$
 x_0 è p.to interno

① $\{f'(x) = 0\}$ possibili p.ti max o min

$$f(x) = x^3 \quad x \in (-1, 1) \quad f'(x) = 3x^2 \quad f'(x) = 0$$



$$f'(0) = 0$$

non è p.to né max né min relat.

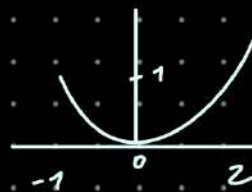
$$3x^2 = 0 \quad x = 0$$

② p.t. f non deriv

③ ESTREMI DEL DOMINIO

$$f(x) = x^2 \quad x \in [-1, 2]$$

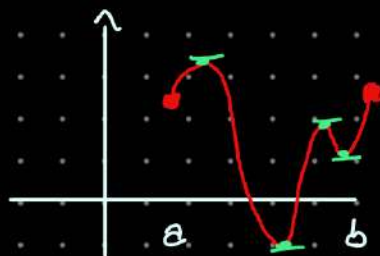
$$f'(x) = 2x \quad f'(x) = 0 \Leftrightarrow x = 0$$



TEOREMA ROLLE

- 1) $f \in C([a, b])$
continua
- 2) f derivabile in (a, b)
- 3) $f(a) = f(b)$

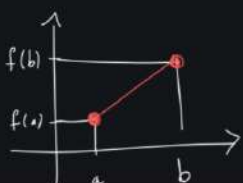
$\Rightarrow \exists c \in (a, b)$ T.C.
 $f'(c) = 0$



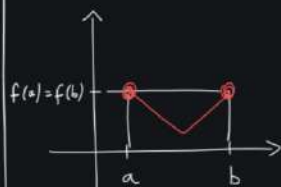
Oss c non necess.
 UNICO

Oss. Le ipotesi di Rolle servono tutte

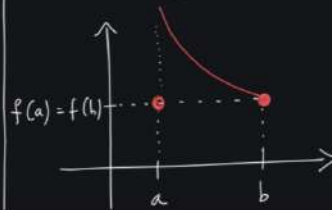
MANCA SOLO
 LA ③



MANCA SOLO
 LA ②



MANCA SOLO
 LA ① (in a)



Dim.

WEIERSTRASS
+
FERMAT

Dato che f è continua e l'intervallo è chiuso $\Rightarrow f$ ha max e min globali (W)

[A] Se almeno uno tra il max e il min glob. cade in (a, b) (DENTRO!) allora possiamo usare Fermat e concludere che in quel punto f' si annulla, a e b

[B] Se il max e il min globali cadono agli estremi, dato che $f(a) = f(b)$, f deve essere costante e quindi $f'(c) = 0$ su tutto (a, b) . $\square$

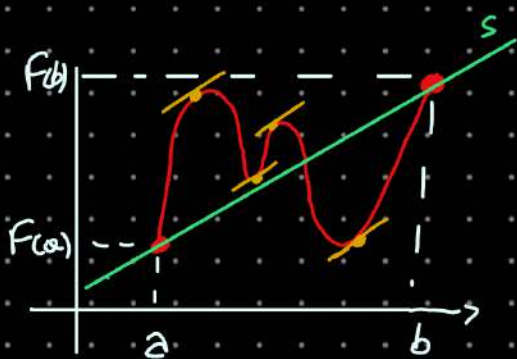
TEOR. LAGRANGE

Sia $f: [a, b] \rightarrow \mathbb{R}$

- 1) f continua su $[a, b]$
- 2) f derivabile su $[a, b]$

$$\Rightarrow \exists c \in (a, b) \text{ t.c. } f'(c) = \frac{f(b) - f(a)}{b - a}$$

OSS



• $\frac{f(b) - f(a)}{b - a}$ è il coeff. angol. della retta S

• $f'(c)$ è il coeff. ang. della tangente in $x=c$

★ OSS

$f \in C(D)$ D chiuso

f derivabile

$$f(a) = f(b)$$

in tutti i pt.
interni a D

a e b il pto più a sx di D $\Rightarrow \exists c: f'(c) = 0$

Derivata II dell'inversa

$(f^{-1})'$ è derivabile $[(f^{-1})'' \text{ esiste}]$ è detta per
 $(f^{-1})''$ —————

Dimostr.

Sappiamo già:

$$(f^{-1})'(y) = \frac{1}{f'(x)} ; y = f(x)$$

$$\begin{aligned} \frac{d^2}{dy^2} (f^{-1}) &= \frac{d}{dy} \left(\frac{d}{dy} f^{-1} \right) = \\ &= \frac{d}{dy} \left(\frac{1}{\underbrace{f'(f^{-1}(y))}} \right) \end{aligned}$$

$$\left[\begin{array}{l} g \\ \bullet z \rightarrow \frac{1}{z} \end{array} \right] \quad \underbrace{g \left[\underbrace{f'}_{\substack{g' \checkmark \\ \text{se } z \neq 0}} \left(\underbrace{f^{-1}(y)} \right) \right]}$$

- f^{-1} è derivab $\forall y$
(per ipotesi f' esiste ed $e' \neq 0$)
- f'' esiste (ipot.)
- g è derivabile in $z \neq 0$

⇓

—————

possiamo usare la chain rule

$$h' = g'(f'(f^{-1})) \cdot (f'(f^{-1}))'$$
$$\underline{g'(f'(f^{-1}))} \cdot f''(f^{-1}) \cdot \underbrace{(f^{-1})'}_{\frac{1}{f'(x)}}$$

$$\bullet g'(z) = -1 z^{-1-1} = -\frac{1}{z^2} \quad z \neq 0$$
$$\frac{1}{f'(x)}$$

$$\bullet g'(f'(f^{-1})) = -\frac{1}{[f'(f^{-1})]}$$

$$(f^{-1})''(y) = -\frac{f''(x)}{[f'(x)]^2}$$

$$(f^{-1})''(y) = \underline{\hspace{10cm}}$$

$$f^{-1}(y) = y^{\frac{1}{\alpha}}$$

$$(f^{-1})'(y) = \frac{1}{\alpha} y^{\frac{1}{\alpha}-1} = \frac{1}{\alpha} y^{\frac{1-\alpha}{\alpha}}$$

$$(f^{-1})''(y) = \frac{1}{\alpha} \left(\frac{1-\alpha}{\alpha} \right) \cdot y^{\frac{1-\alpha}{\alpha}-1}$$

$$\frac{1}{\alpha} \left(\frac{1-\alpha}{\alpha} \right) \cdot y^{\frac{1-2\alpha}{\alpha}}$$

$$= \boxed{\frac{1}{\alpha} \frac{1-\alpha}{\alpha} \cdot x^{1-2\alpha}}$$

Esercizi De L'Hôpital

H1) Dimostrare che il polinomio

$$1+x + \frac{x^2}{2} + \frac{x^3}{6}$$

è un' approssimazione di $f(x) = e^x$ in $x=0$
di un ordine più piccolo di $|x|^3$

in altre parole

$$e^x - p(x) = o(|x|^3)$$

quando $x \rightarrow 0$

anche equival a dire che

$$\lim_{x \rightarrow 0} \frac{e^x - (1+x + \frac{x^2}{2} + \frac{x^3}{6})}{x^3} = 0 \quad \begin{matrix} \nearrow 0 \\ \searrow 0 \end{matrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

• Hôp $\rightarrow \frac{1}{2} \checkmark$

$$\lim_{x \rightarrow 0} \frac{e^x - (1+x + \frac{x^2}{2})}{3x^2} \xrightarrow{\text{h\hat{o}p}} \lim_{x \rightarrow 0} \frac{e^x - (1+x)}{6x} \xrightarrow{\text{h\hat{o}p}} \lim_{x \rightarrow 0} \frac{e^x - 1}{6} = 0 \checkmark$$

$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Analisi limiti forma 1^∞

$$\lim_{x \rightarrow 0} [g(x)]^{f(x)}$$

dove

- $g(x) \rightarrow 1$
 - $f(x) \rightarrow \infty$
- quando $x \rightarrow 0$

Proposizione

il limite "L" esiste $\Leftrightarrow$ il limite

$$L_2 := \lim_{x \rightarrow 0} \frac{\ln(g(x))}{(f(x))^{-1}} \quad \left[\frac{0}{0} \right]$$

- in questo caso $L = e^{L_2}$

↳ Dim

Se L_2 esiste

$$L_2 := \lim_{x \rightarrow 0} \frac{\ln(g(x))}{(f(x))^{-1}} \in \mathbb{R}$$

$$\mathbb{R} \ni e^{L_2} = \lim_{x \rightarrow 0} e^{\left[\frac{\ln(g(x))}{f(x)^{-1}} \right]}$$

$x \leftrightarrow e^z$
 e^z continua

$$= \lim_{x \rightarrow 0} g(x)^{f(x)} = L$$

$e = (\ln)^{-1}$

ESEMPIO

calcolo

$$L = \lim_{x \rightarrow 0} [\cos(x)]^{\frac{1}{x^2}} \quad [1^\infty]$$

- $g(x) = \cos(x)$
- $f(x) = \frac{1}{x^2} \Rightarrow f(x)^{-1} = x^2$
- $\ln(g(x)) = \ln(\cos(x))$

$$L_2 := \lim_{x \rightarrow 0} \frac{\ln(g(x))}{x^2} \quad \left[\frac{0}{0} \right]$$

hóp • $\ln$ deriv in $(0, +\infty)$

• $\cos$ e' deriv

$\rightarrow \ln(g(x))$ e' deriv vicino
a $x=0$

• x^2 e' derivab e $2x \neq 0 \quad \forall x \neq 0$

$$= \lim_{x \rightarrow 0} \frac{[\ln(g(x))]^{-1}}{2x}$$

$$= \lim_{x \rightarrow 0} -\frac{1}{2} \cdot \frac{\tan x}{x} = -\frac{1}{2}$$

$$[\ln(g)]' = \ln'(g) \cdot g'$$

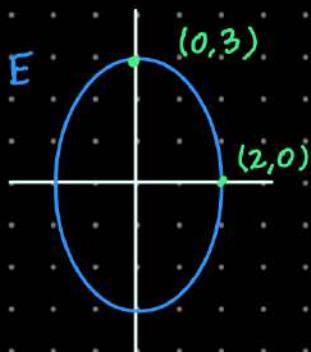
$$= \frac{g'}{g} = -\frac{\sin}{\cos}$$

$$= -\tan$$



$$e^{L_2} = e^{-\frac{1}{2}}$$

Es. dell'ellisse (dato per casa)



• $a, b > 0$

$$ax^2 + by^2 = 1$$

• Se $(2, 0) \in E$

$$\rightarrow a \cdot 4 = 1 \rightarrow a = \frac{1}{4}$$

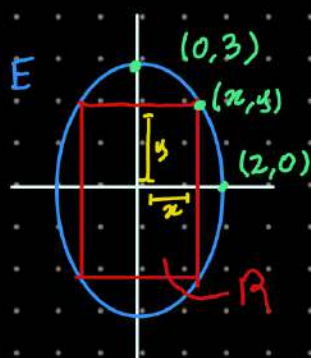
Se $(0, 4) \in E$

$$b = \frac{1}{9}$$

La formula di E e' data per

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

Voglio Trovare il rettangolo con Area massima
che e' inscritto a "E"



$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

$$\Rightarrow y = 3 \sqrt{1 - \frac{x^2}{4}}$$

$$y = \frac{3}{2} \sqrt{4 - x^2}$$

$$\begin{aligned} \text{Area}(R) &= 4xy \\ &= 4x \left(\frac{3}{2}\right) \sqrt{4 - x^2} \\ &= 6x \sqrt{4 - x^2} \end{aligned}$$

Vogliamo massimizzare l'area di $R = R(x)$

-> per semplificare il calcolo, lavoriamo con il quadrato
dell'area

$$\begin{aligned} H(x) &:= \text{Area}(R(x))^2 \\ &= 6^2 x^2 (4 - x^2) \end{aligned}$$

Vogliamo trovare x t.c.

$$H'(x) = 0$$

$$[x^2 (4 - x^2)]' = 0$$

$$2(4 - x^2) = 2x^2$$

$$8 - 2x^2 = 2x^2$$

$$4 = 2x^2$$

$$x = + \sqrt{2}$$

$$(x > 0)$$

$$\left[\lim_{x \rightarrow 0} (\sin^2(x))^{\tan(x)} \right]$$

Domini Fato

~~Domini Fato~~

$$f(x)^{g(x)} : e^{g(x) \ln(f(x))} \quad f(x) > 0$$

$$\frac{d}{dx} (f(x)^{g(x)}) = \frac{d}{dx} (e^{g(x) \ln(f(x))}) = e^{g(x) \ln(f(x))} \left[g'(x) \ln(f(x)) + g(x) \frac{f'(x)}{f(x)} \right]$$

$$\frac{d}{dx} (\sin(x)^{\cos(x)}) = \frac{d}{dx} (e^{\cos(x) \ln(\sin(x))}) = \sin(x)^{\cos(x)} \left[-\sin(x) \ln(\sin(x)) + \cos(x) \frac{\cos(x)}{\sin(x)} \right]$$

$$\bullet \frac{d}{dx} \ln |f(x)| \quad \text{dom}(\ln |f(x)|) = \{x \in \mathbb{R} : f(x) \neq 0\} = \{x \in \mathbb{R} : |f(x)| > 0\}$$

f è derivabile

$$\ln |f(x)| = \begin{cases} \ln(f(x)) & f(x) > 0 \\ \ln(-f(x)) & f(x) < 0 \end{cases}$$

SE $x_0 : f(x_0) > 0$ f continua e per Teor. perm. segno

$$I_\delta = (x_0 - \delta, x_0 + \delta)$$

$$f(x) > 0 \quad \forall x \in I_\delta$$

$$\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)} \quad \Big|_{x=x_0}$$

SE $x_1 : f(x_1) < 0$



$$\bullet \frac{d}{dx} \ln |f(x)| = \frac{f'(x)}{f(x)} \quad \forall x \in D(f), f' \text{ esiste e } f \neq 0$$

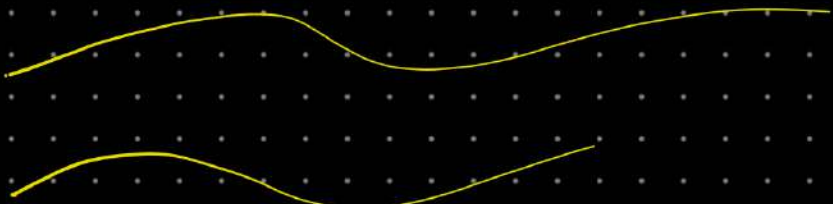
$$\rightarrow \lim_{x \rightarrow 0} \sin^2(x)^{\tan(x)} = \lim_{x \rightarrow 0} e^{\tan(x) \ln(\sin^2(x))} \stackrel{?}{=} \lim_{x \rightarrow 0} e^{\tan(x) 2 \ln \sin x}$$

$\ln(\sin^2(x))$ è ben definito $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ $\setminus \{0\}$ solo se $\sin(x) > 0$ $x \in (0, \frac{\pi}{2})$

$$\ln a^b = b \ln a \quad \text{se } a \geq 0$$

$$0 = \ln(-1)^2 \neq 2 \ln(-1) \quad \text{N.E.}$$

$$\lim_{x \rightarrow 0^+} e^{\tan x} 2 \ln \sin x$$



$$2 \lim_{x \rightarrow 0^+} \tan(x) \ln(\sin(x)) = 2 \lim_{x \rightarrow 0^+} \frac{\tan x}{x} \cdot x^0 \cdot \ln(\sin x)^{-\infty} = 0$$

$$\lim_{x \rightarrow 0^+} x \ln(\sin(x))$$

$$\frac{F}{G} = \frac{\ln(\sin x)}{\frac{1}{x}} \quad \left[\frac{\infty}{\infty} \right]$$

$$\text{L'hop} \quad \frac{F'}{G'} = \frac{\frac{\cos x}{\sin x}}{-\frac{1}{x^2}} = -x^2 \frac{\cos x}{\sin x}$$

$$\lim_{x \rightarrow 0^+} \frac{F'}{G'} = \lim_{x \rightarrow 0^+} \frac{x}{\sin x} \cos x \cdot x = 0$$

hop



$$\bullet \lim_{x \rightarrow +\infty} x \sin(e^{-x})$$

CAMBIO VARIABLE

$$e^{-x} = y \neq 0 \quad (e^{-x} \neq 0)$$

$$-x = \ln y$$

$$\lim_{y \rightarrow 0^+} -\ln y \sin(y) = \lim_{y \rightarrow 0^+} -y \ln(y) \frac{\sin(y)}{y} = 0$$

$$\bullet f(x) = x + x^3 \quad \text{calcola } g'(x) \quad g = f^{-1}$$

① perché f è invertibile?

$$f'(x) = 1 + 3x^2 \geq 1 > 0 \Rightarrow f \text{ è strettam. cresc. su } \mathbb{R}$$

$$y = x + x^3 \quad x = g(y)$$

$$g'(y_0) = \frac{1}{f'(x_0)}$$

se f è derivabile e $f'(x_0) \neq 0$

$$y_0 = f(x_0)$$

$$x_0 = g(y_0)$$

$g'(0)$ DEVO TROVARE x_0 t.c. $f(x_0) = y = 0$

$$x_0 = 0 \quad y_0 = 0 \quad g'(2) \quad y_0 = 2 \quad x_0 = 1$$

$$g'(0) = \frac{1}{f'(0)} = \frac{1}{1+3x^2}|_{x=0} = 1$$

$$g''(0) \quad g'(y_0) = \frac{1}{f'(g(y_0))}$$

$$g''(y) = (g'(y))' = \left(\frac{1}{f'(g(y))} \right)' = - \frac{f''(g(y))}{[f'(g(y))]^2} \quad g'(y) = - \frac{f''(g(y))}{[f'(g(y))]^2} \cdot \frac{1}{f'(g(y))}$$

$$g''(y) = - \frac{f''(g(y))}{(f'(g(y)))^3} \quad g''(y_0) = - \frac{f''(g(y_0))}{[f'(g(y_0))]^3}$$

$$f''(x) = 6x \quad f''(x)|_{x=x_0} = 0$$

• Dim che $x e^{\frac{1}{\sqrt{x}}} > 1 \quad \forall x > 0$

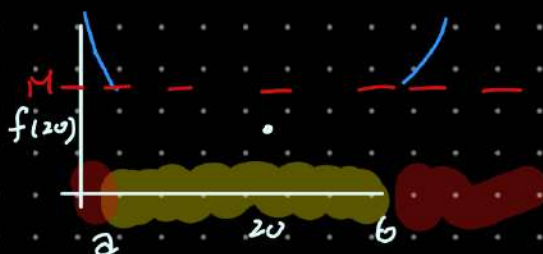
Dom $(0, +\infty)$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x e^{\frac{1}{\sqrt{x}}} \quad \left[\frac{0}{\infty} \right] \quad \text{cambio var}$$

$$\frac{1}{\sqrt{x}} = y \quad \frac{1}{x} = y^2 \quad y = \frac{1}{\sqrt{x}}$$

$$= \lim_{y \rightarrow +\infty} \frac{e^y}{y^2} = +\infty$$

$$\lim_{x \rightarrow +\infty} x e^{\frac{1}{\sqrt{x}}} \rightarrow +\infty$$



① $f \in C(0, +\infty)$ (Dim esistenza minimo)

② $\lim_{x \rightarrow 0^+} f = +\infty \Rightarrow \exists m \in \mathbb{R} \quad m = \min_{(0, +\infty)} f$

③ $\lim_{x \rightarrow +\infty} f = +\infty$



Scelgo un numero a caso $x_0 > 0 \quad f(x_0) > 0$

$$M = f(x_0) + 1$$

② $\exists \delta > 0 \quad \text{t.c.} \quad 0 < x < \delta \quad f(x) > M > f(x_0)$

③ $\exists \kappa \quad \text{t.c.} \quad x > \kappa \quad f(x) > M > f(x_0)$

$$f|_{[\delta, \kappa]} \in C([\delta, \kappa]) \quad \min_{[\delta, \kappa]} f = m$$

$$f(x) > f(x_0) \quad \forall x < \delta \quad \forall x > \kappa$$

$$f(x) = x e^{(x - \frac{1}{2})}$$

$$f'(x) = 1 e^{\frac{1}{\sqrt{x}}} + x e^{\frac{1}{\sqrt{x}}} \left(-\frac{1}{2} x^{-\frac{3}{2}} \right) = e^{\frac{1}{\sqrt{x}}} \left(1 - \frac{1}{2} \frac{x}{\sqrt{x^3}} \right)$$

$$\textcircled{1} f'(x) = 0 \quad (0 < e^{\frac{1}{\sqrt{x}}} \neq 0) \quad \Leftrightarrow \quad 1 - \frac{1}{2\sqrt{x}} = 0 \quad 1 = \frac{1}{2\sqrt{x}}$$

$$\sqrt{x} = \frac{1}{2}$$

$$x = \frac{1}{4}$$

$$f'(x) > 0 \quad \Leftrightarrow \quad 1 - \frac{1}{2\sqrt{x}} > 0 \Rightarrow x > \frac{1}{4}$$



$$m = \min f = f\left(\frac{1}{4}\right)$$

$$f(x) = x e^{\frac{1}{\sqrt{x}}}$$

$$f\left(\frac{1}{4}\right) = \frac{1}{4} e^{\frac{1}{\sqrt{1/4}}} = \frac{1}{4} e^2 > \frac{4}{4} = 1$$

ES

$$(x+y)^2 \leq 2(x^2+y^2)$$

$$\forall x, y \in \mathbb{R}$$

$$(x+y)^2 = x^2 + 2xy + y^2 \leq x^2 + x^2 + y^2 + y^2$$

$$0 \leq (x-y)^2 = x^2 - 2xy + y^2$$

$$\sim \quad 2xy \leq x^2 + y^2$$

Dimostriamo che $p-1$

$$\bullet (x+y)^p \leq 2^{p-1} (x^p + y^p)$$

$$x, y \geq 0$$

$$y=0 \quad x^p \leq 2^{p-1} x^p$$

$$\text{Vera } \forall x \in \mathbb{R}^+$$

$$\frac{(x+y)^p}{y^p} \leq 2^{p-1} \frac{(x^p + y^p)}{y^p}$$

$$\left(\frac{x+y}{y}\right)^p \leq 2^{p-1} \left(\frac{x^p}{y^p} + 1\right)$$

$$\left(\frac{x}{y} + 1\right)^p \leq 2^{p-1} \left(\left(\frac{x}{y}\right)^p + 1\right)$$

$$t = \frac{x}{y} \geq 0$$

$$\hookrightarrow (t+1)^p \leq 2^{p-1} (t^p + 1)$$

$$t \in [0, +\infty)$$

$$\hookrightarrow F'(t) = 2^{p-1} (t^p + 1) - (t+1)^p \geq 0$$

$$F'(t) = 2^{p-1}$$



$$(2t)^{p-1} > (t+1)^{p-1} \quad t > 1$$

f

$\Rightarrow t=1$ è pt. di minimo

$$F(1) = 2^{p-1} (1^p + 1) - (1+1)^p = 2^{p-1} \cdot 2 - 2^p = 0$$

$$F(t) \geq F(1) = 0$$

$$f(x) = \sin(x) \cos(2x) \quad x \in [0, \frac{\pi}{2}]$$

π, n $\exists$ per
Weierstrass

$$\begin{aligned} f'(x) &= \cos(x) \cos(2x) + \sin(x) (-\sin(2x) \cdot 2) \\ &= \cos(x) \cos(2x) - 2 \sin(x) \sin(2x) \\ &= \cos(x) (\cos^2(x) - \sin^2(x)) - 4 \sin^2(x) \cos(x) \\ &= \cos(x) [\cos^2(x) - 5 \sin^2(x)] \\ &= \cos(x) (1 - 6 \sin^2(x)) \end{aligned}$$

$$\cos^2(x) = 1 - \sin^2(x)$$

$$\cos(x) > 0 \quad x \in (0, \frac{\pi}{2})$$

$$1 - 6 \sin^2(x) = 0 \quad \sin^2(x) = \frac{1}{6} \quad \left| \sin(x) = \frac{1}{\sqrt{6}} \quad \left| x = \arcsin \frac{1}{\sqrt{6}} \right. \right.$$

$$x_0 = 0$$

$$x_1 = \arcsin\left(\frac{1}{\sqrt{6}}\right)$$

$$x_2 = \frac{\pi}{2}$$

$$f(x_0) = 0$$

$$f(x_2) = -1$$

$$\begin{aligned} f(x_1) &= \sin(x_1) \cos(2x_1) = \\ &= \frac{1}{\sqrt{6}} (\cos^2(2x_1) - \sin^2(x_1)) \\ &= \frac{1}{\sqrt{6}} \left(1 - \frac{1}{3} \right) = \frac{2}{3\sqrt{6}} \end{aligned}$$

